Loss function – Cost Function

Loss Function is the **error for ONE data point**

Cost Function is the **average error for ALL data points**

Loss function – Cost Function

Example: Linear Regression

Loss (Squared Error):
For 1 data

$$Loss_i = (y_{i,actual} - y_{i,predict})^2$$

Cost(Mean Squared Error – MSE)
For all data

$$Cost = \frac{Loss_1 + Loss_2 + \cdots + Loss_n}{n}$$

x	y_actual	y_predict	Loss
1	2	5	$(2-5)^2 = 9$
4	4	8	$(4-8)^2 = 16$
3	3	9	$(3-9)^2 = 36$

Cost = (9 + 16 + 36)/3
= 20.3

Example of Loss function:

1) Squared Error - SE (Used in Linear regression):

$$Loss = (y - \hat{y})^2$$

Where:

- y = actual value
- $\hat{y}$ = prediction

Cost Function -> **Mean Squared Error - MSE**
It is the average of squared error for all *n* samples

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- Used in house price prediction, salary prediction, etc.
- Penalizes **large errors heavily** (because of square)
- Smooth and differentiable → good for **gradient descent**
- Sensitive to **outliers**

Loss function – Cost Function

NOTE: Some authors define loss function to be same as cost function

That is, loss function = the **average error for ALL data points**

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

2) **Absolute Error** –AE (Used in Linear regression):

$$Loss = |y - \hat{y}|$$

Where:

- y = actual value
- $\hat{y}$ = prediction

Cost Function -> **Mean Absolute Error - MAE**

It is the average of absolute error for all *n* samples

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

x	y_actual	y_predict	Loss
1	2	5	| 2 – 5 |= 3
4	4	8	| 4 – 8 |= 4
3	3	9	| 3 – 9 |= 6

Cost (MAE) = (3 + 4 + 6)/3
= 4.3

Mean Absolute Error - MAE

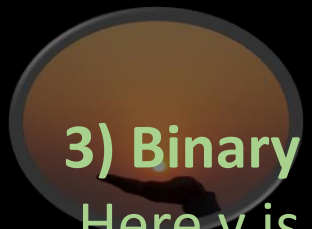
Less sensitive to **outliers,** because there no squaring

$$Loss = |y - \hat{y}|$$

Where:

- y = actual value
- $\hat{y}$ = prediction

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$



3) Binary Cross Entropy (Used in classification):

Here y is 0 or 1.

$$Loss = -(y\log(\hat{y}) + (1-y)\log(1-\hat{y}))$$

Where:

- y = true label (0 or 1)
- $\hat{y}$ = predicted probability of class **1**

Cost Function: It is the average of binary cross entropy for all *n* samples

$$-\frac{1}{n}\sum_{i=1}^{n}[y_i\log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]$$

Use case: Spam detection, Disease prediction, Fraud detection

- Penalizes confident wrong predictions heavily

3) Binary Cross Entropy

Example 1 (Correct Prediction)

$$Loss = -(y\log(\hat{y}) + (1-y)\log(1-\hat{y}))$$

Given

True label

$$y = 1$$

Model predicted probability (of class 1)

$$\hat{y} = 0.9$$

Where:

- y = true label (0 or 1)
- $\hat{y}$ = predicted probability of class 1

Plug into formula

$$Loss = -[1 \cdot \log(0.9) + (1-1)\log(1-0.9)] = 0.105$$

Here this is low loss → good prediction

3) Binary Cross Entropy

Example 2 (Bad Prediction)

$$Loss = -(y\log(\hat{y}) + (1-y)\log(1-\hat{y}))$$

Given

True label

$$y = 1$$

Model predicted probability (of class 1)

$$\hat{y} = 0.1$$

Where:

- y = true label (0 or 1)
- $\hat{y}$ = predicted probability of class 1

Plug into the formula:

$$Loss = -[1\log(0.1) + (1-1)\log(1-0.1)] = 2.302$$

Here this is high loss → bad prediction

Here the model had high confidence for wrong prediction (prob of class 0 = 0.9), BCE penalizes strongly this **confident wrong predictions**

4) Categorical Cross Entropy Loss:

Used in multi-class classification, i.e. there are **more than 2 classes**.

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

Where:

- C = number of classes
- y_i = actual label (one-hot encoded)
- $\hat{y}_i$ = predicted probability

For example, in digit recognition (0 – 9), animal classification (dog-cat-snake)

5) Hinge Loss (Used in classification by SVM):

$$Loss = max(0, 1 - y \cdot \hat{y})$$

Where:

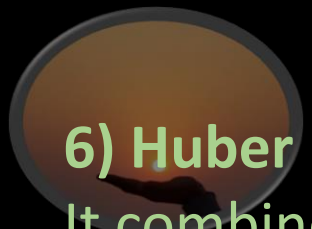
- y = true label (0 or 1)
- $\hat{y}$ = predicted probability of class **1**

Example: If y = 1 and $\hat{y} = 0.3$, then

Loss = max(0, 1 – 1x0.3)

= max(0, 0.7)

= 0.7



6) Huber Loss (Regression):

It combines advantages of **MSE + MAE**.

$$L = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

Where:

- y = actual value
- $\hat{y}$ = prediction
- δ = threshold

7) KL Divergence Loss:

Measures difference between **probability distributions.**

It is used in, VAEs, Generative AI, Knowledge distillation

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)}$$

Where:

- P = true distribution
- Q = predicted distribution

It essentially tells you how much information is lost when using Q instead of P.

STOP

8) Focal Loss:

It is used for **imbalanced classification problems**.

Example

- Fraud detection
- Cancer detection

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

Where:

- p_t = predicted probability of correct class
- α = class weight
- γ = focusing parameter

9) Dice Loss (Image Segmentation):

Measures overlap between prediction and ground truth.

$$Dice = \frac{2|A \cap B|}{|A| + |B|}$$

$$Loss = 1 - Dice$$

Where:

- A = predicted pixels
- B = actual pixels